

Operators in second quantization, Wick's theorems and the particle-hole formalism

FYS4480/FYS9480 – Week 36, September 3–4, 2026

Morten Hjorth-Jensen

Department of Physics and Center for Computing in Science Education,
University of Oslo, Norway

Thursday September 3: reminder of week 35, first and second quantization, one- and two-body operators, Wick's theorem and its proof

Friday September 4: the generalised theorem, applications, the bit representation, and the particle-hole formalism

Reading: these slides, chapter 3 of the lecture notes (sections 3.1–3.4 and 3.11–3.16), Szabo and Ostlund chapter 2, Shavitt and Bartlett chapter 3. Code: `WeeklySlides/week36.ipynb` and `BookPrograms/chapter03/wick.py`

Plan for week 36

Thursday September 3, 2×45 min

- Reminder (10–15 min): the algebra, normal ordering, contractions, two, three and four operators (sections 3.1 and 3.6–3.10)
- From first to second quantization: the dictionary, unitarity, field operators, change of basis (section 3.1)
- The number operator, one-body and two-body operators: the expressions for the rest of the course (sections 3.2–3.3)
- Wick's theorem and its proof by induction (sections 3.11–3.12)

Friday September 4, 2×45 min

- Wick's generalised theorem for products of normal-ordered groups, and its proof (sections 3.13–3.14)
- Applications: one- and two-body matrix elements between two-particle states – the Condon-Slater rules (section 3.15)
- From the formalism to a running code: Slater determinants as bit patterns (section 3.16)
- The particle-hole formalism: the Fermi sea as vacuum, quasiparticle operators, $\hat{H}$ in particle-hole form (section 3.4)
- Exercise session: exercise set for week 36 (chapter 2 of the lecture notes)

Not covered: section 3.17 (the Jordan-Wigner transformation) – we return to it in the quantum-computing part of the course.

Thursday, lecture 1: Reminder, and operators in second quantization

Where week 35 ended

Last week we replaced permutation bookkeeping by operator algebra. What we did:

- defined $a_{\alpha}^{\dagger}$ and a_{α} and wrote every antisymmetric state as a string of creation operators acting on the vacuum (section 3.1);
- **derived** the three anticommutation relations from the antisymmetry of the states;
- defined **normal ordering** and **contractions**, and worked through two, three and four operators by hand (sections 3.6–3.10).

What we did *not* reach, and do this week:

- the number operator and the one- and two-body operators in second quantized form (sections 3.2–3.3),
- Wick's theorem, its generalisation to normal-ordered groups, and the proofs (sections 3.11–3.14),
- the applications, the bit representation and the particle-hole formalism (3.15, 3.16, 3.4).

The next five slides collect last week's results in the form we need them today.

Reminder 1: the algebra

Second quantization for fermions

$$a_{\alpha}^{\dagger}|0\rangle = |\alpha\rangle, \quad |\alpha_1 \dots \alpha_n\rangle = a_{\alpha_1}^{\dagger} \dots a_{\alpha_n}^{\dagger}|0\rangle, \quad a_{\alpha}|0\rangle = 0,$$

$$\{a_{\alpha}^{\dagger}, a_{\beta}^{\dagger}\} = \{a_{\alpha}, a_{\beta}\} = 0, \quad \{a_{\alpha}^{\dagger}, a_{\beta}\} = \delta_{\alpha\beta}$$

- $a_{\alpha}^{\dagger} a_{\beta}^{\dagger} = -a_{\beta}^{\dagger} a_{\alpha}^{\dagger}$ is the antisymmetry of the determinant; $(a_{\alpha}^{\dagger})^2 = 0$ is the Pauli principle.
- a_{α} removes a particle from α ; if α is not occupied the result is zero. Moving α to the front of a ket costs one sign per operator passed.
- The whole art of the subject: **push annihilation operators to the right** with $a_p a_q^{\dagger} = \delta_{pq} - a_q^{\dagger} a_p$, let them die on the vacuum, collect the deltas.

The identity $[AB, C] = A\{B, C\} - \{A, C\}B$ was our workhorse for commutators such as $[\hat{N}, a_{\gamma}^{\dagger}] = a_{\gamma}^{\dagger}$.

Reminder 2: normal ordering

A product is **normal-ordered** when all creation operators stand to the left of all annihilation operators;

$$N[xyz \cdots w] = (-1)^\kappa (\text{same operators, creation operators to the left}),$$

κ the number of interchanges of neighbouring fermion operators needed. For instance

$$N[a_p a_q^\dagger] = -a_q^\dagger a_p, \quad N[a_p^\dagger a_q] = a_p^\dagger a_q, \quad N[a_p a_q^\dagger a_r a_s^\dagger] = a_q^\dagger a_s^\dagger a_p a_r.$$

The one property that matters

$$\langle 0 | N[xyz \cdots w] | 0 \rangle = 0 \quad \text{for every non-empty product,}$$

because a creation operator on the far left or an annihilation operator on the far right kills the vacuum.

Evaluating $\langle 0 | xyz \cdots w | 0 \rangle$ therefore means finding the pieces of the product that are *not* normal-ordered.

Reminder 3: contractions

The **contraction** of two operators is the product minus its normal-ordered product,

$$\overline{xy} \equiv xy - N[xy] = \langle 0|xy|0\rangle,$$

a *number*, which can be pulled outside any operator string. Of the four possible pairs only one survives:

$$\overline{a_p a_q^\dagger} = \delta_{pq},$$

$$\overline{a_p a_q} = 0,$$

$$\overline{a_q^\dagger a_p} = 0,$$

$$\overline{a_p^\dagger a_q^\dagger} = 0.$$

- Only an annihilation operator standing to the *left* of a creation operator contracts to something non-zero: **the vacuum contains no particles**.
- **Crossing rule**: a set of contractions carries the sign $(-1)^\nu$, ν the number of crossings of the contraction lines, one interchange of neighbours per crossing.

Reminder 4: two, three and four operators

- **Two operators.** The definition rearranged: $xy = N[xy] + (\text{contraction})$. With $x = a_p$, $y = a_q^\dagger$ this is $a_p a_q^\dagger = \delta_{pq} - a_q^\dagger a_p$, and $\langle p|q \rangle = \delta_{pq}$.
- **Three operators.** An odd string cannot be paired up completely, so $\langle 0|xyz|0 \rangle = 0$ always. The full expansion of $a_p a_q a_r^\dagger$ reads $N[a_p a_q a_r^\dagger] + \delta_{qr} a_p - \delta_{pr} a_q$ – the second contraction crosses one operator.
- **Four operators.** The overlap of two two-particle states, by anticommutation (four steps) or by inspecting the three pairings (two survive):

$$\langle rs|pq \rangle = \langle 0| \overbrace{a_s a_r} a_p^\dagger a_q^\dagger |0 \rangle + \langle 0| \overbrace{a_s a_r} a_p^\dagger \overbrace{a_q^\dagger} |0 \rangle = \delta_{rp} \delta_{sq} - \delta_{sp} \delta_{rq}.$$

The pattern we extrapolate today

product = normal-ordered product + all single contractions + all double contractions + $\dots$; in a vacuum expectation value only the fully contracted terms survive, each a product of deltas times $(-1)^{\text{crossings}}$.

Pause to think: warming up the machinery

Pause to think #1

(about 3 min – discuss with your neighbour)

1. Read off $\langle 0 | a_p a_q a_r^\dagger a_s^\dagger | 0 \rangle$ directly from the pairings, without any anticommutation. Which of the three pairings vanish, and what sign does each survivor carry?
2. What is $N[a_p a_q^\dagger a_r^\dagger a_s]$? Count the interchanges carefully.

Answer to pause #1

1. The pairing $(a_p a_q)(a_r^\dagger a_s^\dagger)$ vanishes. The nested pairing $(a_p, a_s^\dagger)(a_q, a_r^\dagger)$ has no crossing: $+\delta_{ps}\delta_{qr}$. The crossed pairing $(a_p, a_r^\dagger)(a_q, a_s^\dagger)$ has one crossing: $-\delta_{pr}\delta_{qs}$. Hence $\langle 0|a_p a_q a_r^\dagger a_s^\dagger|0\rangle = \delta_{ps}\delta_{qr} - \delta_{pr}\delta_{qs}$, which is $\langle qp|rs\rangle$ – antisymmetric in (p, q) and in (r, s) , as it must be.
2. Move a_p past $a_q^\dagger$ and $a_r^\dagger$: two interchanges, sign $+$. The annihilation operators a_p, a_s keep their relative order. $N[a_p a_q^\dagger a_r^\dagger a_s] = +a_q^\dagger a_r^\dagger a_p a_s$.

From first to second quantization: the dictionary (section 3.1)

Let $\{\varphi_p(x)\}$ be an orthonormal single-particle basis, x collecting all coordinates of one particle, and $p_1 < p_2 < \dots < p_N$ a set of occupied orbitals. In first quantization the normalised Slater determinant is

$$\Phi_{p_1 \dots p_N}(x_1, \dots, x_N) = \frac{1}{\sqrt{N!}} \det[\varphi_{p_i}(x_j)],$$

in second quantization the same set of orbitals defines the Fock-space vector $a_{p_1}^\dagger \dots a_{p_N}^\dagger |0\rangle$.

The dictionary

$$\Phi_{p_1 \dots p_N}(x_1, \dots, x_N) \longleftrightarrow a_{p_1}^\dagger a_{p_2}^\dagger \dots a_{p_N}^\dagger |0\rangle, \quad p_1 < p_2 < \dots < p_N.$$

- The common ordering convention fixes the sign on both sides: interchanging two labels exchanges two rows of the determinant and two creation operators – both give a minus sign.
- Nothing is ever *computed* when passing from one language to the other. What changes is **where the antisymmetry is kept**: in the $N!$ terms of the wave function, or in the algebra of the operators.

The correspondence is unitary

The map sends one orthonormal basis onto another. On the Fock-space side, pushing the annihilation operators of the bra to the right, every complete pairing of a 's with $a^\dagger$'s contributes a product of deltas with the sign of the pairing read as a permutation:

$$\langle q_1 q_2 \cdots q_N | p_1 p_2 \cdots p_N \rangle = \sum_{\sigma \in S_N} \text{sgn}(\sigma) \prod_{i=1}^N \delta_{q_i p_{\sigma(i)}} = \det[\delta_{q_i p_j}],$$

which for ordered label sets is 1 when the sets coincide and 0 otherwise – orthonormality, on both sides.

Identical coefficients

$$\Psi = \sum_{p_1 < \cdots < p_N} C_{p_1 \cdots p_N} \Phi_{p_1 \cdots p_N} \longleftrightarrow |\Psi\rangle = \sum_{p_1 < \cdots < p_N} C_{p_1 \cdots p_N} a_{p_1}^\dagger \cdots a_{p_N}^\dagger |0\rangle.$$

In one and the same orbital basis the “transformation matrix” between first and second quantization is the identity. A linear map taking an orthonormal basis to an orthonormal basis is unitary, and by linearity the correspondence extends to the whole antisymmetric N -particle space.

Field operators: the isomorphism in closed form

Introduce the **field operators**, which annihilate and create a particle at the point x ,

$$\hat{\psi}(x) = \sum_p \varphi_p(x) a_p, \quad \hat{\psi}^\dagger(x) = \sum_p \varphi_p^*(x) a_p^\dagger, \quad \{\hat{\psi}(x), \hat{\psi}^\dagger(x')\} = \delta(x - x'), \quad \{\hat{\psi}(x), \hat{\psi}(x')\} = 0.$$

Expanding each field operator and using the overlap formula, only the terms in which $(q_1, \dots, q_N)$ is a permutation of $(p_1, \dots, p_N)$ survive – the Leibniz formula for the determinant appears by itself:

$$\langle 0 | \hat{\psi}(x_N) \cdots \hat{\psi}(x_1) a_{p_1}^\dagger \cdots a_{p_N}^\dagger | 0 \rangle = \sum_{\sigma} \text{sgn}(\sigma) \prod_{i=1}^N \varphi_{p_{\sigma(i)}}(x_i) = \det[\varphi_{p_i}(x_j)].$$

Wave function as a matrix element, and its inverse

$$\Psi(x_1, \dots, x_N) = \frac{\langle 0 | \hat{\psi}(x_N) \cdots \hat{\psi}(x_1) | \Psi \rangle}{\sqrt{N!}}, \quad | \Psi \rangle = \frac{1}{\sqrt{N!}} \int dx_1 \cdots dx_N \Psi(x_1, \dots, x_N) \hat{\psi}^\dagger(x_1) \cdots \hat{\psi}^\dagger(x_N) | 0 \rangle$$

The program `slater2fock.py` (chapter 3) checks both formulas on a discretised single-particle space, to machine precision.

Changing the single-particle basis: the exterior power

A genuinely non-trivial transformation matrix appears when two *different* orbital bases are compared. With a unitary U ,

$$\tilde{\varphi}_\alpha(x) = \sum_p \varphi_p(x) U_{p\alpha}, \quad \tilde{a}_\alpha^\dagger = \sum_p U_{p\alpha} a_p^\dagger, \quad \tilde{a}_\alpha = \sum_p U_{p\alpha}^* a_p,$$

the anticommutation relations are preserved precisely because U is unitary, $\{\tilde{a}_\alpha, \tilde{a}_\beta^\dagger\} = \sum_p U_{p\alpha}^* U_{p\beta} = \delta_{\alpha\beta}$. Expanding each $\tilde{a}^\dagger$ and using the overlap formula,

$$\langle p_1 \cdots p_N | \tilde{a}_1 \cdots \tilde{a}_N \rangle = \sum_{\sigma \in S_N} \text{sgn}(\sigma) \prod_{i=1}^N U_{p_{\sigma(i)} \alpha_i} = \det[U_{p_i \alpha_j}].$$

- The transformation matrix between two determinant bases is the matrix of all $N \times N$ minors of U – the N th exterior power $U^{\wedge N}$, a $\binom{n}{N} \times \binom{n}{N}$ unitary matrix.
- If U mixes only the N occupied orbitals among themselves, a single minor is relevant, $\det U$: the rotated determinant equals the original up to a phase (week 34).
- Two answers to one question: in one basis the transformation is the identity; between two bases it is $U^{\wedge N}$. Section 2 of the notebook builds $U^{\wedge N}$ from the Fock-space matrices and checks both statements.

The number operator (section 3.2)

For $\alpha = \beta$ the operator $a_\alpha^\dagger a_\alpha$ counts the occupation of α ,

$$a_\alpha^\dagger a_\alpha |\alpha_1 \alpha_2 \dots \alpha_n\rangle = \begin{cases} 0 & \alpha \notin \{\alpha_i\}, \\ |\alpha_1 \alpha_2 \dots \alpha_n\rangle & \alpha \in \{\alpha_i\}, \end{cases}$$

(the two signs from moving α to the front and back again cancel). Summing over all single-particle states,

$$\hat{N} = \sum_{\alpha} a_\alpha^\dagger a_\alpha, \quad \hat{N} |\alpha_1 \alpha_2 \dots \alpha_n\rangle = n |\alpha_1 \alpha_2 \dots \alpha_n\rangle.$$

- $\hat{N}$ is our first **one-body operator**: it acts on one single-particle state at a time.
- $[\hat{N}, a_\gamma^\dagger] = a_\gamma^\dagger$ and $[\hat{N}, a_\gamma] = -a_\gamma$: creation raises the count by one, annihilation lowers it. Any operator with equally many $a^\dagger$'s and a 's commutes with $\hat{N}$.
- Most systems we study conserve particle number and $[\hat{H}, \hat{N}] = 0$ is a cheap consistency test of a program. In reactions such as (d, p) or (p, d) states with different particle numbers are needed – second quantization handles them for free.

One-body operators: from coordinates to operators

Take $\hat{H}_0 = \sum_i \hat{h}_0(x_i)$ and expand the action of $\hat{h}_0$ in the single-particle basis,

$$\hat{h}_0(x_i)\psi_{\alpha_k}(x_i) = \sum_{\alpha'_k} \psi_{\alpha'_k}(x_i) \langle \alpha'_k | \hat{h}_0 | \alpha_k \rangle.$$

On the product $\psi_{\alpha_1}(x_1) \cdots \psi_{\alpha_n}(x_n)$ the operator replaces, in turn, each orbital α_k by α'_k with weight $\langle \alpha'_k | \hat{h}_0 | \alpha_k \rangle$. Repeating this for every permutation of the determinant and collecting the signs,

$$\hat{H}_0 |\alpha_1 \alpha_2 \dots \alpha_n\rangle = \sum_{k=1}^n \sum_{\alpha'_k} \langle \alpha'_k | \hat{h}_0 | \alpha_k \rangle |\alpha_1 \alpha_2 \dots \alpha'_k \dots \alpha_n\rangle.$$

The replacement $\alpha_k \rightarrow \alpha'_k$ inside a ket is exactly what $a_{\alpha'_k}^\dagger a_{\alpha_k}$ does,

$$|\alpha_1 \alpha_2 \dots \alpha'_k \dots \alpha_n\rangle = a_{\alpha'_k}^\dagger a_{\alpha_k} |\alpha_1 \alpha_2 \dots \alpha_k \dots \alpha_n\rangle,$$

and since $a_{\alpha'}^\dagger a_{\alpha}$ gives zero unless α is occupied, the sum over k can be replaced by a free sum over all single-particle states.

One-body operators: the result, and a three-line check

One-body operator, second quantized form

$$\hat{H}_0 = \sum_{\alpha\beta} \langle \alpha | \hat{h}_0 | \beta \rangle a_{\alpha}^{\dagger} a_{\beta}, \quad \text{more generally} \quad \hat{F} = \sum_{pq} \langle p | \hat{f} | q \rangle a_p^{\dagger} a_q.$$

$\hat{H}_0$ moves a particle from β to α with amplitude $\langle \alpha | \hat{h}_0 | \beta \rangle$; any particle-conserving one-body operator has this form ($\langle \alpha | 1 | \beta \rangle = \delta_{\alpha\beta}$ gives back $\hat{N}$).

Check between single-particle states, by pure algebra:

$$\langle \alpha_1 | \hat{H}_0 | \alpha_2 \rangle = \sum_{\alpha\beta} \langle \alpha | \hat{h}_0 | \beta \rangle \langle 0 | a_{\alpha_1} a_{\alpha}^{\dagger} a_{\beta} a_{\alpha_2}^{\dagger} | 0 \rangle, \quad a_{\alpha_1} a_{\alpha}^{\dagger} a_{\beta} a_{\alpha_2}^{\dagger} = (\delta_{\alpha\alpha_1} - a_{\alpha}^{\dagger} a_{\alpha_1}) (\delta_{\beta\alpha_2} - a_{\alpha_2}^{\dagger} a_{\beta}),$$

and every term still containing an operator dies on the vacuum:

$$\langle 0 | a_{\alpha_1} a_{\alpha}^{\dagger} a_{\beta} a_{\alpha_2}^{\dagger} | 0 \rangle = \delta_{\alpha\alpha_1} \delta_{\beta\alpha_2} \implies \langle \alpha_1 | \hat{H}_0 | \alpha_2 \rangle = \langle \alpha_1 | \hat{h}_0 | \alpha_2 \rangle. \checkmark$$

In Wick language: the two groups a_{α_1} and $a_{\beta} a_{\alpha_2}^{\dagger}$ contract into $a_{\alpha}^{\dagger}$ and $a_{\alpha_2}^{\dagger}$ respectively, with no crossing.

Two-body operators: the result (section 3.3)

Starting from $\hat{H}_I = \sum_{i < j} V(x_i, x_j)$ and proceeding exactly as for $\hat{H}_0$ – expanding the action on products of orbitals, rewriting the replacement of a *pair* of orbitals as

$$a_{\alpha'_k}^\dagger a_{\alpha'_l}^\dagger a_{\alpha_l} a_{\alpha_k} |\alpha_1 \dots \alpha_k \dots \alpha_l \dots \alpha_n\rangle = |\alpha_1 \dots \alpha'_k \dots \alpha'_l \dots \alpha_n\rangle,$$

and lifting the restriction to distinct pairs with the symmetry $\langle \alpha\beta | \hat{v} | \gamma\delta \rangle = \langle \beta\alpha | \hat{v} | \delta\gamma \rangle$ – one finds

Two-body operator, second quantized form

$$\hat{H}_I = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma, \quad \text{all four sums free.}$$

- Note the order of the annihilation operators: $a_\delta a_\gamma$, **reversed** relative to the ket labels of $\langle \alpha\beta | \hat{v} | \gamma\delta \rangle$.
- The derivation (three steps, each analogous to the one-body case) is in section 3.3 of the book; we do not repeat it here.
- The check $\langle \alpha_1 \alpha_2 | \hat{H}_I | \beta_1 \beta_2 \rangle = \langle \alpha_1 \alpha_2 | \hat{v} | \beta_1 \beta_2 \rangle - \langle \alpha_1 \alpha_2 | \hat{v} | \beta_2 \beta_1 \rangle$ is an eight-operator vacuum expectation value: on Friday it becomes a four-term Wick calculation.

Antisymmetrised form, the full Hamiltonian, and a three-body operator

Since the operator string $a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma$ is antisymmetric under $\gamma \leftrightarrow \delta$ we may antisymmetrise the matrix element, at the price of a factor $\frac{1}{2}$:

$$\hat{H}_I = \frac{1}{4} \sum_{\alpha\beta\gamma\delta} [\langle\alpha\beta|\hat{v}|\gamma\delta\rangle - \langle\alpha\beta|\hat{v}|\delta\gamma\rangle] a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma = \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{AS} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma.$$

The prefactor $-\frac{1}{2}$ or $\frac{1}{4}$ – tells which convention is in use.

The Hamiltonian for the rest of the course (antisymmetrised elements assumed)

$$\hat{H} = \sum_{pq} \langle p|\hat{t} + \hat{u}_{\text{ext}}|q\rangle a_p^\dagger a_q + \frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle a_p^\dagger a_q^\dagger a_s a_r.$$

The pattern continues: a three-body force (needed in nuclear physics) is

$$\hat{V}_3 = \frac{1}{36} \sum_{pqrst} \langle pqr|\hat{v}_3|stu\rangle_{AS} a_p^\dagger a_q^\dagger a_r^\dagger a_u a_t a_s, \quad \frac{1}{36} = \frac{1}{(3!)^2},$$

with the fully antisymmetrised three-body element (six terms). An n -body operator carries n creation and n annihilation operators and the prefactor $1/(n!)^2$.

Pause to think: the factors and the operator order

Pause to think #2

(about 4 min – discuss with your neighbour)

1. Why must the annihilation operators appear in the reversed order $a_\delta a_\gamma$ rather than $a_\gamma a_\delta$? (Hint: check the sign of $\langle \gamma\delta | \hat{H}_I | \gamma\delta \rangle$ for two particles in two different orbitals.)
2. Where does the change of prefactor $\frac{1}{2} \rightarrow \frac{1}{4}$ come from when we switch to antisymmetrised matrix elements, and why $\frac{1}{36}$ for three-body forces?

Answer to pause #2

1. With the reversed order, $a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma}$ acting on $|\gamma\delta\rangle = a_{\gamma}^{\dagger} a_{\delta}^{\dagger} |0\rangle$ removes γ first and then δ without picking up an extra transposition, so $\langle\gamma\delta|\hat{v}|\gamma\delta\rangle$ enters the interaction energy with a $+$ sign. With $a_{\gamma} a_{\delta}$ every term changes sign and two repelling particles would attract. (Verified operator by operator in the notebook.)
2. The antisymmetrised element is itself a difference of two terms, and the operator string is antisymmetric under $\gamma \leftrightarrow \delta$ and under $\alpha \leftrightarrow \beta$. Symmetrising the $\frac{1}{2}$ -form counts every distinct contribution twice, so the prefactor drops to $\frac{1}{4}$. For three bodies the string is antisymmetric under all $3!$ permutations of the creation labels and all $3!$ of the annihilation labels; with fully antisymmetrised elements each distinct term is counted $3! \cdot 3! = 36$ times.

The operator dictionary

First quantization	Second quantization	Remark
$\frac{1}{\sqrt{N!}} \det[\varphi_{p_i}(x_j)]$	$a_{p_1}^\dagger \cdots a_{p_N}^\dagger 0\rangle, p_1 < \cdots < p_N$	same coefficients, same signs
$\sum_i 1$	$\hat{N} = \sum_p a_p^\dagger a_p$	counts particles
$\sum_i \hat{f}(x_i)$	$\hat{F} = \sum_{pq} \langle p \hat{f} q\rangle a_p^\dagger a_q$	moves one particle
$\sum_{i<j} \hat{v}(x_i, x_j)$	$\frac{1}{2} \sum_{pqrs} \langle pq \hat{v} rs\rangle a_p^\dagger a_q^\dagger a_s a_r$	plain elements
	$\frac{1}{4} \sum_{pqrs} \langle pq \hat{v} rs\rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r$	antisymmetrised elements
$\sum_{i<j<k} \hat{v}_3$	$\frac{1}{36} \sum \langle pqr \hat{v}_3 stu\rangle_{AS} a_p^\dagger a_q^\dagger a_r^\dagger a_u a_t a_s$	three-body forces
$\varphi_p \rightarrow \tilde{\varphi}_\alpha = \sum_p \varphi_p U_{p\alpha}$	$\tilde{a}_\alpha^\dagger = \sum_p U_{p\alpha} a_p^\dagger$	determinants transform with $U^{\wedge N}$
$\Psi(x_1, \dots, x_N)$	$\frac{1}{\sqrt{N!}} \langle 0 \hat{\psi}(x_N) \cdots \hat{\psi}(x_1) \Psi\rangle$	field operators

Where we stand

Every entry of the right-hand column is a matrix in the notebook, and every identity on these slides is checked there as a matrix identity. The physics sits in the matrix elements $\langle p|\hat{f}|q\rangle$ and $\langle pq|\hat{v}|rs\rangle$, computed once in the chosen basis. What remains is to **evaluate strings of operators** between determinants – and that is Wick's theorem.

Thursday, lecture 2: Wick's theorem and its proof

The pattern – and its limit

Every explicit computation so far followed the same recipe:

1. push annihilation operators to the right with $a_p a_q^\dagger = \delta_{pq} - a_q^\dagger a_p$;
2. let them die on the vacuum, $a_p |0\rangle = 0$;
3. collect the Kronecker deltas that fall out along the way.
 - For two operators: immediate.
 - For the four-operator overlap of two-particle states: an exercise in care.
 - For the eight operators of $\langle \alpha_1 \alpha_2 | \hat{H}_I | \beta_1 \beta_2 \rangle$: half a page in the book (section 3.3).
 - For the **twelve or more** operators of perturbation theory and coupled cluster theory: hopeless – the recursion branches at every step.

Wick's theorem

organises this entire calculation once and for all, in terms of the two objects defined last week: normal-ordered products and contractions.

Wick's theorem (section 3.11)

Wick's theorem

An arbitrary product of creation and annihilation operators equals its normal-ordered product plus the sum of all possible contractions,

$$xyz \cdots w = N[xyz \cdots w] + \underbrace{\sum_{(1)} N[xyz \cdots w]}_{\text{all single contractions}} + \underbrace{\sum_{(2)} N[xyz \cdots w]}_{\text{all double contractions}} + \cdots + \sum_{[M/2]} N[xyz \cdots w].$$

- $\sum_{(m)}$ runs over all ways of choosing m disjoint pairs to contract; $[M/2]$ is the largest number of disjoint pairs that M operators admit.
- $N[\cdots]$ with contractions inside means: the contracted pairs pulled out as numbers, the remaining operators normal-ordered, the whole term carrying the crossing sign $(-1)^\nu$.
- This is an identity between *operators*. Section 5 of the notebook verifies it as such, entry by entry, for random strings of up to eight operators.

Vacuum expectation values: fully contracted terms only

Take the vacuum expectation value of the theorem. Every term that still contains an uncontracted operator is normal-ordered and vanishes. Only the *fully* contracted terms survive:

Wick's theorem for vacuum expectation values

$$\langle 0|xyz \cdots w|0\rangle = \sum_{[M/2]} N[xyz \cdots w] = \sum_{\text{complete pairings}} (-1)^\nu \prod \text{contractions.}$$

- For M even there are $(M-1)!!$ complete pairings – one, three, fifteen, one hundred and five for $M = 2, 4, 6, 8$ – and most vanish on inspection because one of their contractions is of a vanishing type.
- For M odd there are none: odd strings give zero, as established last week.
- Each surviving term is a product of Kronecker deltas times $(-1)^{\text{crossings}}$.

The theorem turns the evaluation of any vacuum expectation value into a **combinatorial enumeration** that can be written down before a single anticommutator is applied.

Counting the work: the two routes compared

For strings of n annihilation followed by n creation operators ($M = 2n$), from the book (Fig. 3.1) and `wick.py`:

M	pairings $(M - 1)!!$	recursive calls	surviving terms $n!$
4	3	9	2
8	105	129	24
10	945	651	120
14	135 135	27 399	5 040

- The theorem does *not* win a race on raw counts – from $M = 10$ the recursion makes fewer calls, because branches that begin with $a^\dagger$ or end with a die at once.
- What separates the routes: the pairings are known **in advance**, a purely combinatorial object; each can be inspected alone, most discarded at a glance, and the $n!$ survivors written down directly with their crossing signs. The recursion delivers one unsimplified answer at the very end and cannot be used as an ingredient in an algebraic argument.
- The ratio $(2n - 1)!!/n!$ – already 27 at $M = 14$ – is what makes hand calculation possible.

Why this matters beyond bookkeeping

Many-body connection

An n -body operator can connect only Slater determinants differing by at most n single-particle states – with a two-body force, at most two. A configuration-interaction (shell-model) Hamiltonian matrix is therefore **extremely sparse**: of the $\binom{n}{N}^2$ possible matrix elements, only a vanishing fraction are non-zero. That sparsity is what iterative eigensolvers such as the Lanczos algorithm exploit, and it is why the Hamiltonian matrix never has to be stored in full. The precise statement – the Condon-Slater rules, this week's exercise 1 – falls out of Wick's theorem in a few lines tomorrow.

Quantum computing connection

On a quantum computer the same strings $a^\dagger a$ and $a^\dagger a^\dagger a a$ become, through the Jordan-Wigner transformation (section 3.17, later in the course), strings of Pauli gates. Wick's theorem is the classical way of evaluating their expectation values; a quantum computer estimates them by repeated measurement instead.

Pause to think: six operators

Pause to think #3

(about 3 min – discuss with your neighbour)

Consider $\langle 0 | a_c a_b a_a a_d^\dagger a_e^\dagger a_f^\dagger | 0 \rangle$ ($M = 6$).

1. How many pairings does Wick's theorem enumerate?
2. How many of them survive the elementary contractions, and why exactly that many?
3. What does each surviving term look like – and what familiar object is the sum?

Answer to pause #3

1. $(6 - 1)!! = 5 \cdot 3 \cdot 1 = 15$.
2. A non-vanishing contraction must join one of the three annihilation operators (all standing left) to one of the three creation operators (all standing right). Every such complete assignment is a permutation of three objects: $3! = 6$ survivors.
3. A product of three deltas with the parity of the permutation as sign, e.g. $+\delta_{ad}\delta_{be}\delta_{cf}$ for the nested pairing and $-\delta_{ae}\delta_{bd}\delta_{cf}$ after one transposition – exactly the six terms of $\det[\delta_{q_i p_j}]$, the overlap formula of section 3.1 for $N = 3$. Wick's theorem knows about determinants without ever being told.

Proof of Wick's theorem: the strategy (section 3.12)

The proof is by induction on the number of operators M .

- The case $M = 2$ is the definition of the contraction rearranged – already established.
- The induction step requires one preparatory result, which does all the real work:

Lemma: appending one operator on the right

Let $x, y, z, \dots, w$ be operators and Ω one further creation or annihilation operator. Then

$$N[xyz \cdots w] \Omega = N[xyz \cdots w \Omega] + \sum_{(1)\Omega} N[xyz \cdots w \Omega],$$

where $\sum_{(1)\Omega}$ runs over all single contractions *involving* Ω – the contractions of Ω with each of $x, y, z, \dots, w$ in turn.

In words: multiplying a normal-ordered product by one more operator costs exactly one round of contractions with that operator.

Proof of the lemma: the three easy cases

- (i) Ω is an annihilation operator. Appending it on the far right cannot disturb the normal ordering, so the left-hand side already equals $N[xyz \cdots w \Omega]$. At the same time every contraction with Ω vanishes: a non-zero contraction needs an annihilation operator to the left of a creation operator, and Ω stands to the right of everything. Both extra terms are absent – the lemma holds trivially.
- (ii) All of $x, y, \dots, w$ are annihilation operators, Ω too. A special case of (i).
- (iii) Ω is a creation operator. Then it suffices to prove the lemma when all of $x, y, \dots, w$ are annihilation operators: any creation operator inside the product already stands to the left of Ω 's eventual position, and its contraction with Ω vanishes,

$$\overline{a_i^\dagger a_j^\dagger} = 0,$$

so creation operators in the string contribute neither to the reordering nor to the contraction sum, and are carried along untouched.

Proof of the lemma: the case that works

- (iv) It remains to move a creation operator Ω leftwards through annihilation operators $x, y, \dots, w$, one at a time. Each step uses

$$a_j \Omega = \delta_{j\Omega} - \Omega a_j,$$

producing two pieces:

- a **Kronecker delta** – precisely the contraction of a_j with Ω , carrying the sign of the operators Ω has crossed so far;
- a term in which Ω has moved **one place to the left**.

Iterating until Ω has passed all of them: the accumulated delta terms are exactly $\sum_{(1)\Omega} N[xyz \cdots w \Omega]$, and the final term – Ω to the left of every annihilation operator – is exactly $N[xyz \cdots w \Omega]$. $\square$

The lemma is nothing but the push-to-the-right recipe, run once and recorded as an identity.

The lemma at work

For $M = 1$:

$$N[a_1] a_\ell^\dagger = N[a_1 a_\ell^\dagger] + N \left[\overbrace{a_1 a_\ell^\dagger} \right] = -a_\ell^\dagger a_1 + \delta_{1\ell},$$

which is $a_1 a_\ell^\dagger = \delta_{1\ell} - a_\ell^\dagger a_1$ once more. For $M = 2$:

$$a_0 N[a_1] a_\ell^\dagger = N[a_0 a_1 a_\ell^\dagger] + N \left[\overbrace{a_0 a_1 a_\ell^\dagger} \right] + N \left[\overbrace{a_0 a_1 a_\ell^\dagger} \right],$$

that is $N[a_0 a_1 a_\ell^\dagger] + \delta_{1\ell} N[a_0] - \delta_{0\ell} N[a_1]$, the last sign from carrying the contraction past a_1 – last week's three-operator expansion, now derived rather than guessed. Generally,

$$a_0 N[a_1 a_2 \cdots a_M] a_\ell^\dagger = N[a_0 a_1 \cdots a_M a_\ell^\dagger] + \sum_{(1)\Omega} N[a_0 a_1 \cdots a_M a_\ell^\dagger].$$

The induction step

Assume the theorem holds for M operators. Multiply both sides from the right by Ω and apply the lemma to every term:

- the leading term gives $N[xyz \cdots w] \Omega = N[xyz \cdots w \Omega] + \sum_{(1)\Omega} N[xyz \cdots w \Omega]$;
- each m -fold contracted term is a number times a normal-ordered product of the uncontracted operators, so the lemma applies to it too:

$$\sum_{(m)} N[xyz \cdots w] \Omega = \sum_{(m) \neq \Omega} N[xyz \cdots w \Omega] + \sum_{(m)\Omega} N[xyz \cdots w \Omega],$$

splitting the m -fold contractions of $M + 1$ operators into those not involving Ω and those in which one contraction ends on Ω .

But an m -fold contraction of the $M + 1$ operators *either involves Ω or it does not*, so the two sums recombine into $\sum_{(m)}$ over all $M + 1$ operators. Collecting terms by the number of contractions reproduces the theorem with $M \rightarrow M + 1$. $\square$

A remark on parity, and where we stand

If $M + 1$ is odd, the deepest sum still leaves one operator uncontracted; on taking the vacuum expectation value those terms vanish, and we recover: *vacuum expectation values of odd strings are zero*. For $M + 1$ even the deepest sum consists of the fully contracted terms, and the boxed vacuum-expectation-value formula follows.

What we now own

An algorithm – enumerate pairings, discard vanishing contractions, attach crossing signs – that evaluates *any* product of creation and annihilation operators, with a proof that it equals the anticommutation route, and a program (`wick.py`) that checks both.

Many-body connection

In practice we never meet a bare product. We meet $\langle \Phi | \hat{O} | \Phi' \rangle$: a bra, an operator and a ket, each *already* normal-ordered. Tomorrow's generalised theorem says that contractions *inside* such a group can be skipped – that is what cuts the 105 pairings of an eight-operator string down to 24, and the 24 down to the four that contribute.

Pause to think: where was Ω 's nature used?

Pause to think #4

(about 3 min – discuss with your neighbour)

1. In the induction step we multiplied by Ω *on the right*. Why is it enough to prove the theorem this way – what about products that grow to the left?
2. Which single ingredient of the whole proof would fail for *bosons*, and what would replace it?

Answer to pause #4

1. Every product of $M + 1$ operators *is* a product of its first M operators times its last one, so right-multiplication reaches all strings; induction starting from $M = 2$ covers them all. (One can equally run the induction leftwards – the lemma has a mirror image with Ω prepended.)
2. The sign bookkeeping: for bosons the elementary relation is a *commutator*, $[b_p, b_q^\dagger] = \delta_{pq}$, interchanges cost no sign, and the crossing rule becomes trivial (+1 always). Wick's theorem holds for bosons too, with contractions $\langle 0|b_p b_q^\dagger|0\rangle = \delta_{pq}$ and no signs – the fermionic $(-1)^\nu$ is the only difference.

Summary of Thursday, and what comes on Friday

Today

- Reminder: the algebra, normal ordering, contractions, the crossing rule
- Section 3.1: the dictionary between determinants and operator strings is the identity in one basis and U^N between two bases; field operators
- $\hat{N}, \hat{F} = \sum \langle p | \hat{f} | q \rangle a_p^\dagger a_q,$
 $\hat{V} = \frac{1}{4} \sum \langle pq | \hat{v} | rs \rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r$
- Wick's theorem; proof by one lemma plus induction

Friday

- The generalised theorem: products of normal-ordered groups, and its proof
- Applications: $\langle ij | \hat{O}^{(1)} | k \ell \rangle$ and $\langle ij | \hat{H}_I | ij \rangle$ – the Condon-Slater rules
- Slater determinants as bit patterns: the formalism as code
- The particle-hole formalism: the Fermi sea as vacuum
- Exercise session (exercises 1 and 2 of the week-36 set)

Before Friday: run sections 1–5 of `week36.ipynb`, and redo pause #1 with the string $a_p a_q^\dagger a_r a_s^\dagger$ – which pairings survive now that the operators alternate?

Friday, lecture 3: The generalised theorem, applications, and the bit representation

Where we are

Thursday gave us the operators,

$$\hat{F} = \sum_{pq} \langle p | \hat{f} | q \rangle a_p^\dagger a_q, \quad \hat{V} = \frac{1}{4} \sum_{pqrs} \langle pq | \hat{v} | rs \rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r,$$

and Wick's theorem for an arbitrary product of operators.

Today:

- what we actually meet is $\langle \Phi | \hat{O} | \Phi' \rangle$ – a product of **normal-ordered groups**. Wick's generalised theorem (sections 3.13–3.14) says that contractions inside a group may be skipped;
- the two calculations that motivated the whole construction: $\langle ij | \hat{O}^{(1)} | k\ell \rangle$ and $\langle ij | \hat{H}_I | ij \rangle$ – the Condon-Slater rules by contractions (section 3.15);
- how all of this becomes code: Slater determinants as bit patterns (section 3.16);
- and in the last 45 minutes: the Fermi sea as a new vacuum – the particle-hole formalism (section 3.4) – and the exercises.

Wick's generalised theorem (section 3.13)

Products of normal-ordered groups

$$N[A_1 A_2 \cdots] N[B_1 B_2 \cdots] N[C_1 C_2 \cdots] = N[A_1 A_2 \cdots B_1 B_2 \cdots C_1 C_2 \cdots] + \sum_{\text{all}} N[A_1 A_2 \cdots C_1 C_2 \cdots],$$

where the sum runs over all contractions *between operators belonging to different groups*.

- Contractions within a single group are absent – not thrown away, but **zero**, because that group is already normal-ordered and its internal contractions are of the vanishing types.
- The ordinary theorem is the special case in which every group holds a single operator.
- Taking the vacuum expectation value:

$$\langle 0 | N[A_1 A_2 \cdots] N[B_1 B_2 \cdots] N[C_1 C_2 \cdots] | 0 \rangle = \sum_{\substack{\text{fully contracted,} \\ \text{no line inside a group}}} (-1)^\nu \prod \text{contractions}$$

the form used relentlessly in Hartree-Fock, perturbation theory and coupled cluster theory.

What the restriction buys

Put numbers on it before proving anything (all from `wick.py`, and recomputed in section 6 of the notebook):

string	groups	all pairings	inter-group pairings
one-body element $\langle 0 a_j a_i a_p^\dagger a_q a_k^\dagger a_\ell^\dagger 0 \rangle$	$2 + 2 + 2$	$5!! = 15$	8
two-body element $\langle 0 a_j a_i a_p^\dagger a_q^\dagger a_s a_r a_i^\dagger a_j^\dagger 0 \rangle$	$2 + 4 + 2$	$7!! = 105$	24
three two-body operators	$4 + 4 + 4$	$11!! = 10\,395$	1 728

- Cutting a string into k groups of equal size does *not* tame the factorial (book, Fig. 3.2): the surviving fraction settles almost at once on a number fixed by the group size – a little under 0.6 for groups of two, about 0.18 for groups of four.
- **Fat groups pay:** a matrix element is not $2n$ separate operators but a few normal-ordered blocks, and the bigger the blocks the fewer pairings survive – before a single index has been looked at.
- This is why one always writes the interaction in normal-ordered form *before* starting a derivation, never after.

Proof, step 0: a normal-ordered group has no internal contractions (section 3.14)

Throughout, each group is *written* in normal-ordered form (creation operators to the left). This costs nothing: any group can be brought to that arrangement at the price of a sign $(-1)^{\kappa}$ which multiplies both sides of the theorem and cancels. Then $N[A_1 A_2 \cdots]$ is the product $A_1 A_2 \cdots$.

Pick two operators A_k, A_l with $k < l$ of such a group. Two cases:

- A_k is a creation operator: A_l is either a creation operator ($a^\dagger a^\dagger$ contracts to zero) or an annihilation operator standing to its right ($a^\dagger a$ contracts to zero).
- A_k is an annihilation operator: everything to its right is an annihilation operator too (aa contracts to zero).

$$\overbrace{A_k \cdots A_l} = 0 \quad \text{for any two operators of the same normal-ordered group.}$$

The arrangement matters. Written the other way round, the pair $a_p^\dagger a_q$ becomes $a_q a_p^\dagger$, whose contraction is δ_{pq} and not zero at all – and the bookkeeping would have to be redone.

Proof, step 1: two groups

We first prove

$$N[A_1 \cdots A_n] N[B_1 \cdots B_m] = N[A_1 \cdots A_n B_1 \cdots B_m] + \sum_{A-B} N[A_1 \cdots A_n B_1 \cdots B_m],$$

each line joining an operator of the first group to one of the second, by induction on m .

- $m = 1$: the appending lemma of Thursday, verbatim, with $\Omega = B_1$ – every contraction it produces joins the two groups.
- $m \rightarrow m + 1$: multiply the statement for m by B_{m+1} on the right. Every term on the right-hand side is a number times a *single* normal-ordered product of the still uncontracted operators, so the lemma applies to each term separately with $\Omega = B_{m+1}$ and yields
 - (a) the term in which B_{m+1} is absorbed into the normal-ordered product,
 - (b) the contractions of B_{m+1} with the uncontracted operators inside the bracket.

In (b) a partner from the first group gives a legitimate $A-B$ line. A partner from $B_1, \dots, B_m$ gives a line internal to the second group, and it **vanishes by step 0**, since $B_1 \cdots B_{m+1}$ is normal-ordered and B_{m+1} stands to the right of all of them.

Proof, step 1 (continued) and step 2: any number of groups

Collecting the survivors. Every set of A – B contractions of the $n + m + 1$ operators appears exactly once: either it leaves B_{m+1} uncontracted (case (a) applied to a term of the induction hypothesis) or it contracts B_{m+1} with some A_k (case (b)). Each carries the crossing sign, because the lemma appends Ω from the right and the sign it generates is precisely the number of operators the new line crosses. $\square$

Step 2. A second induction, on the number of groups. Suppose the theorem holds for g groups and multiply from the right by one more, $N[H_1 \cdots H_k]$. Each term on the left is again a number times a single normal-ordered product, so step 1 applies with A standing for what is left of the first g groups and B for the new one: it produces the uncontracted term and all contractions between H and the remaining operators – all inter-group lines – and no contractions internal to H .

Every admissible set of contractions of the $g + 1$ groups is generated exactly once, because it splits uniquely into the lines that touch H and those that do not. $\square$

Nothing new was needed

The whole proof is the appending lemma applied over and over, plus step 0. Section 6 of the notebook verifies the generalised theorem as an operator identity for two, three and four groups.

Pause to think: the generalised theorem at work

Pause to think #5

(about 4 min – discuss with your neighbour)

1. For the string $\langle 0 | a_j a_i a_p^\dagger a_q a_k^\dagger a_\ell^\dagger | 0 \rangle$ with the three groups indicated, list the pairings that join only different groups. There should be eight. How many of them survive the elementary contractions?
2. A colleague writes the one-body operator as $\sum_{pq} \langle p | \hat{f} | q \rangle a_q a_p^\dagger$ (“it is the same up to a sign and a constant”) and applies the generalised theorem with this as the middle group. What goes wrong?

Answer to pause #5

1. Label positions 1..6: $a_j, a_i \mid a_p^\dagger, a_q \mid a_k^\dagger, a_\ell^\dagger$. Inter-group pairings: $\{(1, 3), (2, 5), (4, 6)\}$, $\{(1, 3), (2, 6), (4, 5)\}$, $\{(1, 4), (2, 5), (3, 6)\}$, $\{(1, 4), (2, 6), (3, 5)\}$, $\{(1, 5), (2, 3), (4, 6)\}$, $\{(1, 5), (2, 4), (3, 6)\}$, $\{(1, 6), (2, 3), (4, 5)\}$, $\{(1, 6), (2, 4), (3, 5)\}$ – eight. A survivor needs every line to join an annihilation operator $(1, 2, 4)$ to a creation operator $(3, 5, 6)$ standing to its right, which kills every pairing containing $(1, 4)$, $(2, 4)$, $(3, 5)$ or $(3, 6)$. Exactly *four* survive: $\{(1, 3), (2, 5), (4, 6)\}$, $\{(1, 3), (2, 6), (4, 5)\}$, $\{(1, 5), (2, 3), (4, 6)\}$, $\{(1, 6), (2, 3), (4, 5)\}$ – one for each of the four terms of the result on the next slide.
2. The middle group is then *not* normal-ordered: its internal contraction $a_q a_p^\dagger \rightarrow \delta_{pq}$ is not zero, and the theorem as stated omits a non-vanishing term ($\sum_p \langle p | \hat{f} | p \rangle = \text{tr } \hat{f}$ times the overlap). Either bring the group to normal order first, $a_q a_p^\dagger = \delta_{pq} - a_p^\dagger a_q$, or include the internal contraction by hand. Step 0 is a hypothesis, not a decoration.

Application 1: a one-body operator between two-particle states

With $\hat{O}^{(1)} = \sum_{pq} \langle p | \hat{O}^{(1)} | q \rangle a_p^\dagger a_q$ and $|ij\rangle = a_i^\dagger a_j^\dagger |0\rangle$, $\langle ij| = \langle 0 | a_j a_i$,

$$\langle ij | \hat{O}^{(1)} | k\ell \rangle = \sum_{pq} \langle p | \hat{O}^{(1)} | q \rangle \langle 0 | \underbrace{a_j a_i}_{\text{bra}} \underbrace{a_p^\dagger a_q}_{\text{operator}} \underbrace{a_k^\dagger a_\ell^\dagger}_{\text{ket}} | 0 \rangle.$$

Only contractions between different groups contribute (section 3.15), and each must join an annihilation operator to a creation operator standing to its right. One such term:

$$\begin{array}{c} \boxed{} \quad \boxed{} \\ a_j a_i a_p^\dagger a_q a_k^\dagger a_\ell^\dagger \\ \boxed{} \end{array} = + \delta_{ip} \delta_{qk} \delta_{j\ell} \longrightarrow \delta_{j\ell} \langle i | \hat{O}^{(1)} | k \rangle$$

after the sums over p and q are carried out. No line crosses another: sign $+$.

Application 1: the result – the Condon-Slater rules

Collecting the four surviving terms (pause #5),

One-body operator between two-particle determinants

$$\langle ij | \hat{O}^{(1)} | k\ell \rangle = \delta_{j\ell} \langle i | \hat{o}^{(1)} | k \rangle - \delta_{jk} \langle i | \hat{o}^{(1)} | \ell \rangle - \delta_{i\ell} \langle j | \hat{o}^{(1)} | k \rangle + \delta_{ik} \langle j | \hat{o}^{(1)} | \ell \rangle.$$

- Diagonal, $k = i, \ell = j$: $\langle i | \hat{o} | i \rangle + \langle j | \hat{o} | j \rangle$ – the sum over occupied orbitals of week 34.
- One orbital replaced, $|k\ell\rangle = |i\ell'\rangle$ with $\ell' \neq j$: a single term, $\langle j | \hat{o} | \ell' \rangle$ (up to the sign of bringing the determinants to maximum coincidence).
- Two orbitals replaced: no assignment of the deltas can be satisfied – **zero**.

These are the **Condon-Slater rules** for one-body operators (this week's exercise 1a–c), obtained without a single explicit permutation. The same argument for N particles: the bra and ket supply N annihilation and N creation operators, and one contraction must pass through the operator.

Application 2: the two-body interaction, diagonal element

With $\hat{H}_I = \frac{1}{2} \sum_{pqrs} \langle pq|v|rs \rangle a_p^\dagger a_q^\dagger a_s a_r$ (plain elements),

$$\langle ij|\hat{H}_I|ij \rangle = \frac{1}{2} \sum_{pqrs} \langle pq|v|rs \rangle \langle 0| \underbrace{a_j a_i}_{\text{bra}} \underbrace{a_p^\dagger a_q^\dagger a_s a_r}_{\text{operator}} \underbrace{a_i^\dagger a_j^\dagger}_{\text{ket}} |0 \rangle.$$

An eight-operator string: 105 pairings, 24 between different groups, **four** survive. Two of them:

$$\overbrace{a_j a_i a_p^\dagger a_q^\dagger} \overbrace{a_s a_r a_i^\dagger a_j^\dagger} = \delta_{ip} \delta_{jq} \delta_{ri} \delta_{sj} \longrightarrow \langle ij|v|ij \rangle,$$

and the one obtained by exchanging the roles of r and s (the lines $(a_s, a_i^\dagger)$ and $(a_r, a_j^\dagger)$), which crosses one more line and enters with the opposite sign: $-\langle ij|v|ji \rangle$. The two remaining terms are these two with $i \leftrightarrow j$ in the operator indices; they are equal to the first two and cancel the factor $\frac{1}{2}$.

Application 2: the result

Two-body interaction in a two-particle determinant

$$\langle ij | \hat{H}_I | ij \rangle = \langle ij | v | ij \rangle - \langle ij | v | ji \rangle = \langle ij | v | ij \rangle_{AS}.$$

- Exactly the direct-minus-exchange structure that chapter 2 (week 34) obtained by writing out the antisymmetriser and sorting through permutations. The two derivations agree, as they must.
- This one required no permutations at all – only the observation that a contraction is non-zero when an annihilation operator stands to the left of a creation operator, and that lines inside a group can be ignored.
- For N particles the same four-term pattern appears for every occupied pair, giving $\frac{1}{2} \sum_{ij} \langle ij | v | ij \rangle_{AS}$: the energy functional of week 34, and exercise 1a of this week.

wick.py and section 7 of the notebook perform every calculation of these two slides three ways – brute-force anticommutation, the ordinary theorem and the generalised theorem – and find them identical; they also list which of the (p, q, r, s) assignments survive.

Pause to think: off-diagonal elements

Pause to think #6

(about 4 min – discuss with your neighbour)

Use the generalised theorem to evaluate, for four distinct orbitals i, j, k, ℓ ,

$$\langle k\ell | \hat{H}_I | ij \rangle = \frac{1}{2} \sum_{pqrs} \langle pq | v | rs \rangle \langle 0 | a_\ell a_k a_p^\dagger a_q^\dagger a_s a_r a_i^\dagger a_j^\dagger | 0 \rangle.$$

1. How many inter-group pairings survive, and what is the result?
2. What happens if the determinants differ in only *one* orbital, $\langle kj | \hat{H}_I | ij \rangle$ – and, for N particles, if they differ in three?

Answer to pause #6

1. The bra operators must contract into $a_p^\dagger a_q^\dagger$ (two ways) and $a_s a_r$ into the ket (two ways): four survivors, exactly as for the diagonal element. Result:
 $\langle k\ell | \hat{H}_I | ij \rangle = \langle k\ell | v | ij \rangle - \langle k\ell | v | ji \rangle = \langle k\ell | v | ij \rangle_{AS}$ – a two-body operator connects determinants differing by two orbitals with a single antisymmetrised matrix element (exercise 1c).
2. With one orbital in common the four survivors give $\langle kj | v | ij \rangle_{AS}$; for N particles the common orbital can be any occupied m , and the result is $\sum_m \langle km | v | im \rangle_{AS}$ – an *effective one-body* operator, the interaction part of the Fock matrix (exercise 1b). With three or more differing orbitals at least one bra operator has no creation operator to contract with except those of the ket, and the two-body operator cannot bridge the gap: **zero**. The Hamiltonian matrix is sparse.

From the formalism to a running code (section 3.16)

The time-consuming part of any shell-model or FCI code is the action of the Hamiltonian on a determinant,

$$\left(\sum_{\alpha\beta} \langle \alpha | t + u | \beta \rangle a_{\alpha}^{\dagger} a_{\beta} + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle a_{\alpha}^{\dagger} a_{\beta}^{\dagger} a_{\delta} a_{\gamma} \right) a_{\alpha_1}^{\dagger} a_{\alpha_2}^{\dagger} \cdots a_{\alpha_n}^{\dagger} |0\rangle.$$

A practically useful way to implement it: **encode a Slater determinant as a bit pattern**. With n single-particle states $\alpha_0, \dots, \alpha_{n-1}$ and $N \leq n$ particles, an occupied state is a 1, an empty one a 0. For $n = 16$ and particles in $\alpha_3, \alpha_6, \alpha_{10}, \alpha_{13}$:

$$\Phi_{3,6,10,13} = a_{\alpha_3}^{\dagger} a_{\alpha_6}^{\dagger} a_{\alpha_{10}}^{\dagger} a_{\alpha_{13}}^{\dagger} |0\rangle = |0001001000100100\rangle, \quad 2^3 + 2^6 + 2^{10} + 2^{13} = 9288.$$

- The ordering of the states is important: it defines the order of the creation operators, and hence the sign convention – the dictionary of section 3.1 with a fixed ordering.
- There are 2^n bit patterns but only $\binom{n}{N}$ determinants with N particles: $\dim \mathcal{H} = \binom{n}{N} = 1820$ for $n = 16$, $N = 4$.
- The Fock-space matrices in the notebook are this representation: the row and column indices are the bit patterns.

The action of a creation operator on a bit pattern

$$a_{\alpha_4}^\dagger \Phi_{3,6,10,13} = a_{\alpha_4}^\dagger a_{\alpha_3}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = -a_{\alpha_3}^\dagger a_{\alpha_4}^\dagger a_{\alpha_6}^\dagger a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = -|0001101000100100\rangle,$$

one anticommutation to put $a_{\alpha_4}^\dagger$ in its place, while $a_{\alpha_6}^\dagger \Phi_{3,6,10,13} = -a_{\alpha_3}^\dagger (a_{\alpha_6}^\dagger)^2 a_{\alpha_{10}}^\dagger a_{\alpha_{13}}^\dagger |0\rangle = 0$.

The recipe

- If bit b_j is already 1 and we act with $a_{\alpha_j}^\dagger$: return the null vector (Pauli).
- If $b_j = 0$: set it to 1 and return the sign factor $(-1)^l$, l the number of bits set *before* bit j (the creation operators that $a_{\alpha_j}^\dagger$ has to pass).

For a_{α_j} : return zero if $b_j = 0$; otherwise clear the bit and return $(-1)^l$ with the same l .

$a_{\alpha_2}^\dagger 00111\rangle$	$=$	0	$a_{\alpha_2}^\dagger 10101\rangle$	$=$	0
$a_{\alpha_2}^\dagger 01011\rangle$	$=$	$(-1) 01111\rangle$	$a_{\alpha_2}^\dagger 10110\rangle$	$=$	0
$a_{\alpha_2}^\dagger 01101\rangle$	$=$	0	$a_{\alpha_2}^\dagger 11001\rangle$	$=$	$(+1) 11101\rangle$
$a_{\alpha_2}^\dagger 10011\rangle$	$=$	$(-1) 10111\rangle$	$a_{\alpha_2}^\dagger 11010\rangle$	$=$	$(+1) 11110\rangle$

(bits written $\alpha_0\alpha_1\alpha_2\alpha_3\alpha_4$ from left to right, as in the book).

The phase without a loop: bit masks

Consider $a_{\alpha_4}^\dagger a_{\alpha_0} \Phi_{0,3,6,10,13}$. Removing α_0 is a logical AND with the complement of the word $S_1 = |1000 \dots\rangle$; adding α_4 is a logical OR with $S_2 = |00001000 \dots\rangle$, after checking that the AND of S_2 with the state is zero. The phase of the pair move $a_{\alpha_j}^\dagger a_{\alpha_i}$ is

$$(-1)^{\#\{\text{occupied states strictly between } \alpha_i \text{ and } \alpha_j\}},$$

because the sign from removing α_i and the sign from inserting α_j share all the occupied states outside the interval, which therefore cancel in pairs. The mask of the states between α_i and α_j is obtained from S_1 and S_2 by integer arithmetic, the AND with the state selects the occupied ones, and a **bit-count** (popcount) gives l – here $|0001000 \dots\rangle$, $l = 1$, phase -1 .

```
def move(state, i, j): # a_j^dagger a_i acting on a bit pattern
    if not (state >> i) & 1: return 0, None # i empty: null vector
    if i == j: return 1, state # a_i^dagger a_i counts the occupation
    if (state >> j) & 1: return 0, None # j occupied: null vector
    lo, hi = min(i, j), max(i, j)
    between = state & ((1 << hi) - (1 << (lo + 1))) # occupied states strictly between
    phase = -1 if bin(between).count("1") % 2 else 1
    return phase, (state & ~(1 << i)) | (1 << j)
```

Section 8 of the notebook applies this to every pattern of a small space and compares with the Fock-space matrices: they agree bit for bit, sign for sign. This is the inner loop of every large-scale FCI code.

Summary of lecture 3, and what comes next

- The generalised theorem: for products of normal-ordered groups only **inter-group** contractions appear; the proof is the appending lemma twice over.
- $\langle ij|\hat{O}^{(1)}|k\ell\rangle$: four terms, the Condon-Slater rules for one-body operators.
 $\langle ij|\hat{H}_I|ij\rangle = \langle ij|v|ij\rangle_{AS}$: the direct and exchange terms of week 34, from four surviving pairings out of 105.
- An n -body operator connects determinants differing by at most n orbitals – the sparsity that makes large-scale diagonalisation possible.
- Bit patterns: the algebra of $a^\dagger$ and a is a handful of bit operations and a popcount.

Many-body connection

So far the vacuum was empty and every contraction of the type $a^\dagger a$ vanished. In the last 45 minutes we make the **filled Fermi sea** the vacuum. Then $a_i^\dagger a_j$ with i, j below the Fermi level contracts to δ_{ij} as well, the Hamiltonian splits into pieces that create and destroy particle-hole pairs, and expectation values in the reference determinant become one-line computations.

Friday, lecture 4: The particle-hole formalism, and exercises

No free lunch – and one important exception (section 3.4)

Second quantization is an elegant formalism for constructing states and operators, and expectation values are easy to compute. But from a practical point of view – *solving* the Schrödinger equation – there is no particular gain: the many-body problem is equally hard in any representation.

There is at least one case where second quantization comes to our rescue. It is easy to introduce **another reference state than the empty vacuum**: with many particles present we take a filled determinant

$$|c\rangle \equiv |\alpha_1 \alpha_2 \dots \alpha_{n-1} \alpha_n\rangle = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_n}^\dagger |0\rangle \quad (c \text{ for core})$$

as the new vacuum and describe everything else as *excitations* of it.

- This reduces the dimensionality of the many-body problem considerably, and allows us to sum specific many-body correlations to infinite order (perturbation theory, coupled cluster).
- The states $|c\rangle$, $a_{\alpha_{n+1}}^\dagger |c\rangle$, $a_{\alpha_n} |c\rangle$, $a_{\alpha_{n+1}}^\dagger a_{\alpha_n} |c\rangle$, $\dots$ are the reference state, one-particle, one-hole and one-particle-one-hole states.

The reference state and its neighbours

In the original representation the $n + 1$ - and $n - 1$ -particle states are products of $a^\dagger$'s on $|0\rangle$. Relative to $|c\rangle$,

$$|\alpha_1 \alpha_2 \dots \alpha_n \alpha_{n+1}\rangle = a_{\alpha_1}^\dagger \dots a_{\alpha_n}^\dagger a_{\alpha_{n+1}}^\dagger |0\rangle = (-1)^n a_{\alpha_{n+1}}^\dagger |c\rangle \equiv (-1)^n |\alpha_{n+1}\rangle_c,$$

$$|\alpha_1 \alpha_2 \dots \alpha_{n-1}\rangle = a_{\alpha_1}^\dagger \dots a_{\alpha_{n-1}}^\dagger |0\rangle = (-1)^{n-1} a_{\alpha_n} |c\rangle \equiv (-1)^{n-1} |\alpha_n^{-1}\rangle_c.$$

- The first is a **one-particle state**: one particle added to the reference. The second is a **one-hole state**: one particle removed. The signs are the cost of moving $a_{\alpha_{n+1}}^\dagger$ (or a_{α_n}) past the n (or $n - 1$) creation operators of the core.
- The crucial difference from the true vacuum:

$$a_\alpha |c\rangle \neq 0 \quad \text{for } \alpha \in \{\alpha_1, \dots, \alpha_n\}, \quad \text{whereas} \quad a_\alpha |0\rangle = 0 \quad \text{for all } \alpha.$$

An annihilation operator no longer kills the vacuum – so the operators must be redefined.

Quasiparticle operators

We want new operators $b_\alpha^\dagger, b_\alpha$ that behave with respect to $|c\rangle$ exactly as $a^\dagger, a$ did with respect to $|0\rangle$:

$$b_\alpha|c\rangle = 0, \quad \{b_\alpha^\dagger, b_\beta^\dagger\} = \{b_\alpha, b_\beta\} = 0, \quad \{b_\alpha^\dagger, b_\beta\} = \delta_{\alpha\beta}, \quad \langle c|c\rangle = 1.$$

With F the **Fermi level** – the last occupied orbit of $|c\rangle$ – the definition is

Quasiparticle creation and annihilation operators

$$b_\alpha^\dagger = \begin{cases} a_\alpha^\dagger & \alpha > F \\ a_\alpha & \alpha \leq F \end{cases} \quad b_\alpha = \begin{cases} a_\alpha & \alpha > F \\ a_\alpha^\dagger & \alpha \leq F \end{cases} \quad b_\alpha^\dagger|c\rangle = \begin{cases} a_\alpha^\dagger|c\rangle = |\alpha\rangle & \alpha > F \\ a_\alpha|c\rangle = |\alpha^{-1}\rangle & \alpha \leq F \end{cases}$$

- Above F nothing changes. Below F the roles are swapped: *removing* a particle from the core creates a **hole**, and a hole is a quasiparticle.
- The interpretation is that of quasiparticle states: a state obtained by adding one quasiparticle to $|c\rangle$ need not be one particle coupled to a core – it may be one particle missing from it.

Pause to think: the quasiparticle algebra

Pause to think #7

(about 4 min – discuss with your neighbour)

1. Show, from the definition and the a -algebra, that $\{b_\alpha^\dagger, b_\beta\} = \delta_{\alpha\beta}$ for all four combinations of α, β above or below F , and that $b_\alpha|c\rangle = 0$ for every α .
2. What are $b_i^\dagger b_i|c\rangle$ and $b_i b_i^\dagger|c\rangle$ for a hole state $i \leq F$? Compare with $a_i^\dagger a_i|c\rangle$.

Answer to pause #7

1. Both above F : $\{a_a^\dagger, a_b\} = \delta_{ab}$. Both below: $\{b_i^\dagger, b_j\} = \{a_i, a_j^\dagger\} = \delta_{ij}$. Mixed: $\{b_a^\dagger, b_i\} = \{a_a^\dagger, a_i^\dagger\} = 0$ and $\{b_i^\dagger, b_a\} = \{a_i, a_a\} = 0$ – and $\delta_{ai} = 0$ anyway. The b 's inherit the fermion algebra because swapping $a \leftrightarrow a^\dagger$ for a subset of the labels preserves all three relations. For the vacuum: $b_a|c\rangle = a_a|c\rangle = 0$ since a is empty, and $b_i|c\rangle = a_i^\dagger|c\rangle = 0$ since i is occupied (Pauli).
2. $b_i^\dagger b_i|c\rangle = a_i a_i^\dagger|c\rangle = 0$: no quasiparticle in the state i . $b_i b_i^\dagger|c\rangle = a_i^\dagger a_i|c\rangle = |c\rangle$. So $b_i^\dagger b_i$ counts *holes* while $a_i^\dagger a_i = 1 - b_i^\dagger b_i$ counts *particles* in an orbital below F – the origin of the minus sign in the number operator on the next slide.

Particle-hole states, and two number operators

With the new operators we construct many-body quasiparticle states – one-particle-one-hole, two-particle-two-hole, ... – exactly as we constructed many-particle states before:

$$|\beta_1\beta_2\ldots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\ldots\gamma_{n_h}^{-1}\rangle \equiv \underbrace{b_{\beta_1}^\dagger b_{\beta_2}^\dagger \cdots b_{\beta_{n_p}}^\dagger}_{>F} \underbrace{b_{\gamma_1}^\dagger b_{\gamma_2}^\dagger \cdots b_{\gamma_{n_h}}^\dagger}_{\leq F} |c\rangle.$$

The number operator, rewritten with $a_i^\dagger a_i = b_i b_i^\dagger = 1 - b_i^\dagger b_i$ for $i \leq F$,

$$\hat{N} = \sum_{\alpha} a_{\alpha}^\dagger a_{\alpha} = \sum_{\alpha > F} b_{\alpha}^\dagger b_{\alpha} + n_c - \sum_{\alpha \leq F} b_{\alpha}^\dagger b_{\alpha}, \quad \hat{N}|\beta_1 \ldots \gamma_{n_h}^{-1}\rangle = (n_p + n_c - n_h)|\beta_1 \ldots \gamma_{n_h}^{-1}\rangle,$$

with n_c the number of particles in $|c\rangle$: $\hat{N}$ counts **particles**. The operator

$$\hat{N}_{qp} = \sum_{\alpha} b_{\alpha}^\dagger b_{\alpha}, \quad \hat{N}_{qp}|\beta_1 \ldots \gamma_{n_h}^{-1}\rangle = (n_p + n_h)|\beta_1 \ldots \gamma_{n_h}^{-1}\rangle,$$

counts **quasiparticles** – particles above F plus holes below. For a particle-conserving Hamiltonian, $n_p = n_h$ in every state it connects to $|c\rangle$.

The one-body operator in particle-hole form

We split $\hat{H}_0 = \sum_{\alpha\beta} \langle \alpha | \hat{h}_0 | \beta \rangle a_\alpha^\dagger a_\beta$ according to whether α, β lie above or below F , substitute the b 's, and bring each term to normal order with respect to the b 's (only the hole-hole term needs one anticommutation, $b_i b_j^\dagger = \delta_{ij} - b_j^\dagger b_i$):

$\hat{H}_0$, with the quantum-chemistry labels $a, b > F$ (particles) and $i, j \leq F$ (holes)

$$\hat{H}_0 = \sum_{ab} \langle a | \hat{h}_0 | b \rangle b_a^\dagger b_b + \sum_{ai} \left[\langle a | \hat{h}_0 | i \rangle b_a^\dagger b_i^\dagger + \langle i | \hat{h}_0 | a \rangle b_i b_a \right] + \sum_i \langle i | \hat{h}_0 | i \rangle - \sum_{ij} \langle j | \hat{h}_0 | i \rangle b_i^\dagger b_j.$$

- First term: acts on particle states only. Last term: on hole states only, with a **minus sign** – a hole in a deep orbital costs energy $-\langle i | \hat{h}_0 | i \rangle$.
- Second term: **creates or destroys a particle-hole pair** – the only piece that connects $|c\rangle$ to other states.
- Third term: a number, the contribution of the core $|c\rangle$ itself.

From here on the labels carry the constraints: $i, j, k, \dots \leq F$, $a, b, c, \dots > F$, $p, q, r, \dots$ anything, and the $\leq F$, $> F$ on the sums are dropped.

The two-body operator in particle-hole form: five groups

Translating the four indices of $\hat{H}_I = \frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r$ into particle and hole labels, and normal-ordering with respect to the b 's, one can regroup $\hat{H}_I = \hat{H}_I^{(a)} + \hat{H}_I^{(b)} + \hat{H}_I^{(c)} + \hat{H}_I^{(d)} + \hat{H}_I^{(e)}$ by the number of particle indices (the notebook, section 9, verifies the whole decomposition as a matrix identity):

$$\hat{H}_I^{(a)} = \frac{1}{4} \sum_{abcd} \langle ab|\hat{v}|cd\rangle b_a^\dagger b_b^\dagger b_d b_c \quad \text{particle-particle scattering}$$

$$\hat{H}_I^{(b)} = \frac{1}{2} \sum_{abci} \left[\langle ab|\hat{v}|ci\rangle b_a^\dagger b_b^\dagger b_i^\dagger b_c + \langle ci|\hat{v}|ab\rangle b_c^\dagger b_i b_b b_a \right] \quad 1p \rightarrow 2p1h \text{ and back}$$

$$\hat{H}_I^{(c)} = \frac{1}{4} \sum_{abij} \left[\langle ab|\hat{v}|ij\rangle b_a^\dagger b_b^\dagger b_j^\dagger b_i^\dagger + \langle ij|\hat{v}|ab\rangle b_i b_j b_b b_a \right] \quad 2p2h \text{ creation/destruction}$$

$$+ \sum_{abij} \langle ai|\hat{v}|bj\rangle b_a^\dagger b_j^\dagger b_b b_i + \sum_{abi} \langle ai|\hat{v}|bi\rangle b_a^\dagger b_b \quad \text{ph scattering; particle energy from core}$$

Two more groups on the next slide. Each term **conserves the number of particles** and changes the number of quasiparticles by $-4, -2, 0, +2$ or $+4$.

The two-body operator in particle-hole form: the last two groups

$$\hat{H}_I^{(d)} = \frac{1}{2} \sum_{aijk} \left[\langle ai|\hat{v}|jk\rangle b_a^\dagger b_k^\dagger b_j^\dagger b_i + \langle jk|\hat{v}|ai\rangle b_i^\dagger b_j b_k b_a \right] \quad 1h \rightarrow 1p2h \text{ and back}$$

$$+ \sum_{aij} \langle ai|\hat{v}|ji\rangle \left[b_a^\dagger b_j^\dagger + b_j b_a \right] \quad \text{ph creation/destruction by the core}$$

$$\hat{H}_I^{(e)} = \frac{1}{4} \sum_{ijkl} \langle kl|\hat{v}|ij\rangle b_i^\dagger b_j^\dagger b_l b_k - \sum_{ijk} \langle ij|\hat{v}|kj\rangle b_k^\dagger b_i + \frac{1}{2} \sum_{ij} \langle ij|\hat{v}|ij\rangle \quad \text{hh scattering; hole energy; core energy}$$

- The pieces with two $b^\dagger$'s and two b 's scatter quasiparticles; those with four $b^\dagger$'s or four b 's create or destroy two pairs; those with three-plus-one change one quasiparticle into three.
- The **one-body** pieces, $\sum_{abi} \langle ai|\hat{v}|bi\rangle b_a^\dagger b_b$, $-\sum_{ijk} \langle ij|\hat{v}|kj\rangle b_k^\dagger b_i$ and $\sum_{aij} \langle ai|\hat{v}|ji\rangle (b_a^\dagger b_j^\dagger + b_j b_a)$, are the interaction of one quasiparticle with all the particles of the core: together with $\hat{h}_0$ they form the **Fock operator** $f_{pq} = \langle p|\hat{h}_0|q\rangle + \sum_i \langle pi|\hat{v}|qi\rangle$ of Hartree-Fock theory.
- The prefactors follow from counting how many of the (p, q) and (r, s) assignments of $\frac{1}{4} \sum_{pqrs}$ land in each group: $\frac{1}{4}$ when both labels of a pair are of the same kind, $\frac{1}{2}$ when one pair is mixed, 1 when both are.

Pause to think: the energy of the reference state

Pause to think #8

(about 3 min – discuss with your neighbour)

1. Read off $\langle c | \hat{H} | c \rangle = \langle c | \hat{H}_0 + \hat{H}_I | c \rangle$ from the particle-hole forms of the last three slides, without computing anything. Which terms can contribute?
2. Where have you seen the result before?

Answer to pause #8

1. Every term containing at least one b or $b^\dagger$ is normal-ordered with respect to $|c\rangle$ and has vanishing expectation value – the same argument as for $|0\rangle$, now with $b_\alpha|c\rangle = 0$ and $\langle c|b_\alpha^\dagger = 0$. Only the two constants survive:

$$E_{\text{ref}} = \langle c|\hat{H}|c\rangle = \sum_i \langle i|\hat{h}_0|i\rangle + \frac{1}{2} \sum_{ij} \langle ij|\hat{v}|ij\rangle_{\text{AS}}.$$

2. It is the energy functional of week 34 – the one-body sum over occupied orbitals plus the direct-minus-exchange two-body sum – and exercise 1a of this week. Obtained there from $N!$ permutations, here by reading off the terms with no operators. Minimising it over the orbitals gives Hartree-Fock (later in the course).

Wick's theorem relative to the Fermi vacuum

Nothing in today's proofs used the fact that $|0\rangle$ is the true vacuum. Replace $|0\rangle \rightarrow |c\rangle$ and $a_p, a_p^\dagger \rightarrow b_p, b_p^\dagger$: normal ordering, contractions, both theorems and their proofs carry over word for word. Only the elementary contractions change. In terms of the a 's, **two** survive:

$$\overline{a_a a_b^\dagger} = \delta_{ab} \quad (a, b > F), \quad \overline{a_i^\dagger a_j} = \delta_{ij} \quad (i, j \leq F),$$

the second with no counterpart above: a hole state is occupied in $|c\rangle$, so removing and replacing a particle there gives unity.

- Normal ordering relative to $|c\rangle$: quasiparticle creation operators ($a_a^\dagger, a_i$) to the left, quasiparticle annihilation operators ($a_a, a_i^\dagger$) to the right. The particle-hole forms of $\hat{H}_0$ and $\hat{H}_I$ above are precisely $\hat{H}$ expanded in such normal-ordered products, the constants being the full contractions.

Many-body connection

Later in the course we write this compactly, $\hat{H} = E_{\text{ref}} + \hat{F}_N + \hat{V}_N$ (section 3.5), and use the Fermi-vacuum theorem to derive Hartree-Fock theory. In perturbation theory and coupled cluster theory the two contractions above become the particle and hole *lines* of Goldstone diagrams.

Exercise session: week 36

The two exercises of the week-36 set (chapter 2 of the lecture notes, “Exercise session, week 36”, with full answers) are precisely this week’s applications, done once with determinants and once with operators.

Exercise 1: Condon-Slater rules for expectation values

Consider three N -particle Slater determinants $|SD\rangle$, $|SD_i^j\rangle$ and $|SD_{ij}^{kl}\rangle$, where $|SD_i^j\rangle$ differs from $|SD\rangle$ by a single single-particle state – ψ_i replaced by ψ_j – and $|SD_{ij}^{kl}\rangle$ differs by two. Define, as in week 35, a general one-body operator $\hat{F} = \sum_i^N \hat{f}(x_i)$ and a general two-body operator $\hat{G} = \sum_{i>j}^N \hat{g}(x_i, x_j)$ with $\hat{g}$ invariant under interchange of two coordinates. The single-particle states are not necessarily eigenstates of $\hat{f}$.

- Find $\langle SD|\hat{F}|SD\rangle$ and $\langle SD|\hat{G}|SD\rangle$.
- Find $\langle SD|\hat{F}|SD_i^j\rangle$ and $\langle SD|\hat{G}|SD_i^j\rangle$.
- Find $\langle SD|\hat{F}|SD_{ij}^{kl}\rangle$ and $\langle SD|\hat{G}|SD_{ij}^{kl}\rangle$. What happens to the two-body operator for a transition $\langle SD|\hat{G}|SD_{ijk}^{lmn}\rangle$, in which the determinant on the right differs by more than two single-particle states?

Hints: bring the two determinants to *maximum coincidence* first (common orbitals in the same order), otherwise every result acquires the sign of the permutation needed. Then either repeat the permutation argument of week 34, or – recommended – translate to $\langle 0|\cdots|0\rangle$ and use the generalised theorem as in pause #6.

Exercise session: week 36 (continued)

Exercise 2: a first encounter with second quantisation

- a) Show that the density of particles at x ,

$$n(x) = N \int dx_2 \cdots dx_N |\Psi_{AS}(x, x_2, \dots, x_N)|^2, \quad \text{can be written as} \quad n(x) = \sum_k |\psi_k(x)|^2.$$

- b) Calculate the matrix elements $\langle \alpha_1 \alpha_2 | \hat{F} | \alpha_1 \alpha_2 \rangle$ and $\langle \alpha_1 \alpha_2 | \hat{G} | \alpha_1 \alpha_2 \rangle$ with

$$|\alpha_1 \alpha_2\rangle = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger |0\rangle, \quad \hat{F} = \sum_{\alpha\beta} \langle \alpha | f | \beta \rangle a_\alpha^\dagger a_\beta, \quad \hat{G} = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta | g | \gamma\delta \rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma,$$

where the matrix elements are the usual one-body, direct and exchange integrals. Compare with exercise 1c) of week 35.

- 2a) is the first-quantized version of a one-body expectation value: $n(x)$ is $\langle \sum_i \delta(x - x_i) \rangle$, and in second quantization it is $\langle \Phi | \hat{\psi}^\dagger(x) \hat{\psi}(x) | \Phi \rangle$ with the field operators of lecture 1 – try both routes. Section 10 of the notebook checks the identity numerically for three particles in a harmonic-oscillator basis.
- 2b) is application 1 and application 2 of lecture 3, with $\hat{F}$ and $\hat{G}$ in place of $\hat{O}^{(1)}$ and $\hat{H}_I$: two-plus-two-plus-two and two-plus-four-plus-two operators, four surviving pairings each.

What to do with the notebook

WeeklySlides/week36.ipynb contains, in this order:

1. Fermion operators as matrices in a finite Fock space; the algebra verified (reminder of week 35).
2. Section 3.1: the overlap $\det[\delta_{q_i p_j}]$, and the transformation matrix between two determinant bases as the matrix of minors $U^{\wedge N}$, verified to be unitary.
3. $\hat{N}$, $\hat{H}_0$ and $\hat{H}_I$ as matrices: eigenvalues, $[\hat{N}, \hat{H}] = 0$, the $\frac{1}{2}$ - and $\frac{1}{4}$ -forms, the operator-order sign of pause #2, $\langle \alpha_1 \alpha_2 | \hat{H}_I | \beta_1 \beta_2 \rangle$.
4. Normal ordering and the elementary contractions as matrix identities (reminder).
5. Wick's theorem as an *operator* identity – all partial contractions with their signs – for random strings; the cost table.
6. The generalised theorem as an operator identity for two, three and four groups; the pairing counts $15 \rightarrow 8$, $105 \rightarrow 24$, $10\,395 \rightarrow 1\,728$.
7. Applications: the four-term one-body formula and $\langle ij | \hat{H}_I | k\ell \rangle$, three ways, with the surviving assignments listed.
8. Bit patterns: `move(state,i,j)` against the matrices for every state; $\hat{H}|\Phi\rangle$ by bit operations.
9. Particle-hole formalism: the b operators, $\hat{N}$ and $\hat{N}_{qp}$, the particle-hole forms of $\hat{H}_0$ and of the five groups of $\hat{H}_I$ checked as matrix identities, E_{ref} , the two Fermi-vacuum contractions.
10. The exercises: Condon-Slater rules verified for random $\hat{f}$, $\hat{g}$ and $N = 3$; the density $n(x)$ of three oscillator states by brute-force integration against $\sum_k |\psi_k(x)|^2$.

Play with it: change the number of orbitals, the Fermi level, the operator strings.

Summary of week 36

The formalism

- Determinant $\leftrightarrow$ operator string: the identity in one basis, $U^{\wedge N}$ between two.
- $\hat{F} = \sum \langle p|\hat{f}|q\rangle a_p^\dagger a_q$,
 $\hat{V} = \frac{1}{4} \sum \langle pq|\hat{v}|rs\rangle_{\text{AS}} a_p^\dagger a_q^\dagger a_s a_r$.
- Fermi sea as vacuum: $b_a^\dagger = a_a^\dagger$, $b_i^\dagger = a_i$; $\hat{H}$ splits into constants, one-body and pair-creating pieces.
- $E_{\text{ref}} = \sum_i \langle i|\hat{h}_0|i\rangle + \frac{1}{2} \sum_{ij} \langle ij|\hat{v}|ij\rangle_{\text{AS}}$.

The theorems

- Wick: product = normal-ordered + all contractions, signs from crossings; proof by the appending lemma and induction.
- Generalised: for normal-ordered groups only inter-group lines; $105 \rightarrow 24 \rightarrow 4$ for a two-body matrix element.
- Condon-Slater: an n -body operator connects determinants differing by at most n orbitals.
- Relative to $|c\rangle$: $a_a a_b^\dagger \rightarrow \delta_{ab}$ and $a_i^\dagger a_j \rightarrow \delta_{ij}$.

In code: a Slater determinant is an integer, $a^\dagger$ and a are bit operations, the sign a popcount.

Next week, and a closing remark

Next week (week 37)

Full configuration interaction (FCI) theory: Slater determinants as a basis, the configuration-interaction expansion, the structure of the Hamiltonian matrix (the Condon-Slater rules of today decide which elements are non-zero), the exponential wall, and FCI for the pairing model – the three-level model of week 35 grown to full size. Reading: the FCI chapter of the lecture notes (chapter 5) and section 3.5 for the normal-ordered Hamiltonian. Hartree-Fock theory follows after FCI.

Two weeks ago every expectation value cost a page of permutations. Today $\langle c | \hat{H} | c \rangle$ was read off in one line, $\langle k\ell | \hat{H}_I | ij \rangle$ in four surviving pairings, and the whole apparatus fits in a hundred lines of Python. The remaining chapters of the course – configuration interaction, Hartree-Fock, perturbation theory, coupled cluster, and their quantum-computing counterparts – are applications of this week's two theorems to ever longer strings of operators; in FCI the bit representation of section 3.16 is the code that evaluates them.